CS315 · ALGORITHMS ANALYSIS & DESIGN

Midterm 451 Practice

30/10/2023 · 1.5 hours · 25 marks

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01 About this midterm

The midterm of semester 451, dated 30/10/2023, by Prof. Dr. Mohammed Al-Hagery. It lasts 1.5 hours and is marked out of 25. The questions are copied from a photo of the paper.

Question	1	2	3	4	5	6	Total
Marks	4	4	2	5	4	6	25

Important

I have no official answer key for this midterm. The solutions are my own, and every table was computed by code. The matrix chain, LCS and B-tree questions come from lectures whose slides I have not received, so I solved them with the standard textbook method. If the instructor's slides do it differently, follow the slides.

02 Questions

Q1 MATRIX CHAIN · 4 PTS

Apply the Dynamic programming algorithm and give the optimal solution to the matrix chain multiplication ABCD where the order for matrix.

A = 4 × 5, B = 5 × 8, C = 8 × 3, D = 3 × 7.

Note: Provide the best order of matrix multiplication and the number of operations.

Minimum operations = _____ , Best order of multiplication: _____

	A	B	C	D
A				
B				
C				
D				

Q2 HASH TABLE · 4 PTS

(A): Insert integer keys A = [4, 13, 49, 6, 40, 33] in order into a hash table of size 9 using the hash functions h1(k) = k mod 9. Collisions should be resolved via chaining, where collisions are stored at the end of a chain. Fill in the hash table after all keys have been inserted. [2 pts]

(B): Collisions should be resolved via double hashing, and $h_2(k) = 1 + (k \bmod 5)$. Fill in the hash table after all keys have been inserted. [2 pts]

0	1	2	3	4	5	6	7	8

0	1	2	3	4	5	6	7	8

Q3 COUNTING OPERATIONS · 2 PTS

Count the total primitive operations of the following algorithm.

```
Algorithm arrayMax(A, n)
   currentMax ← A[0]
   Temp ← A[0]
   for i ← 1 to n − 1 do
      if A[i] > currentMax then
         currentMax ← A[i]
   { increment counter i }
   return currentMax
```

Q4 TRUE / FALSE + MCQ · 5 PTS

Choose the correct answer by selecting True or False:

1) The code length in Huffman codes depend on the frequency of occurrence of characters: a. True b. False

2) Greedy algorithm can solve all the dynamic programming problems. a. True b. False

3) An algorithm which uses the past results and uses them to find new results is a. Greedy Method b. Divide and conquer c. Dynamic programming d. None of the mentioned

4) The time to run MERGESORT on a list of size n, $T(n) = T(\frac{n}{2}) + T(\frac{n}{2}) + T(\frac{MERGE}{2})$ a. True b. False

5) The recursive activity-selection problem algorithm takes a time of O(n log(n)) a. O(n log(n)) b. $O(n^2)$ c. Other value

Q5 LCS · 4 PTS

Using the dynamic programming approach, find the longest common subsequence between two strings X = ACBAED and Y = ABCADF.

	⌀	A	B	C	A	D	F
⌀							
A							
C							
B							
A							
E							
D							

Q6 B-TREE · 6 PTS

a) Create B-Tree of order 5 from the following list of data items: 25, 53, 91, 41, 12, 87, 62, 51 [1 pts]

b) What will the tree look like if we insert key 30? [1 pt]

c) What will the tree look like if we delete keys 53 and 12 from the tree. [2 pts]

d) delete D and then delete X [2 pts]

